



UNIVERSITY OF
BIRMINGHAM

Superconductivity

Lecture 20 – “Excited above the Gap”

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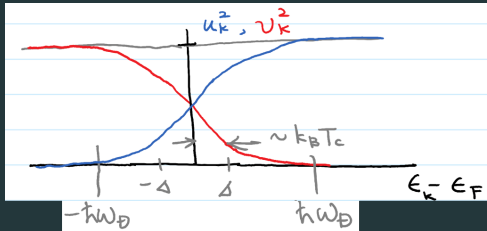
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Previously, BCS wavefunction

$$|BCS\rangle = \prod_{\mathbf{k}} \left(u_{\mathbf{k}} + v_{\mathbf{k}} c_{\mathbf{k}\uparrow}^{\dagger} c_{-\mathbf{k}\downarrow}^{\dagger} \right) |0\rangle, \quad (1)$$

$$\mathcal{H} = \sum_{\mathbf{k}, \sigma} \epsilon_{\mathbf{k}} n_{\mathbf{k}\sigma} - |V| \sum_{\mathbf{k}\mathbf{k}'} c_{\mathbf{k}\uparrow}^{\dagger} c_{-\mathbf{k}\downarrow}^{\dagger} c_{-\mathbf{k}'\downarrow} c_{\mathbf{k}'\uparrow} \quad (2)$$

$$\Delta \simeq 2\hbar\omega_D e^{-1/D(E_F)V} \quad (3)$$



Ground state energy

$$\begin{aligned}\langle E \rangle_s &= \langle BCS | \mathcal{H} - \mu N_{op} | BCS \rangle = \sum_{\mathbf{k}} \left[\left(\epsilon_{\mathbf{k}} - \mu \right) - \frac{(\epsilon_{\mathbf{k}} - \mu)^2}{E_{\mathbf{k}}} \right] - \frac{\Delta_{\mathbf{k}}^2}{|V|} \\ &= \sum_{\mathbf{k}} \left[\left(\epsilon_{\mathbf{k}} - \frac{(\epsilon_{\mathbf{k}} - \mu)^2}{E_{\mathbf{k}}} \right) - \frac{\Delta_{\mathbf{k}}^2}{2E_{\mathbf{k}}} \right]\end{aligned}$$

where $E_{\mathbf{k}} = \sqrt{(\epsilon_{\mathbf{k}} - \mu)^2 + \Delta_{\mathbf{k}}^2}$

$$\langle E \rangle_n = 2 \sum_{|\mathbf{k}| < k_F} (\epsilon_{\mathbf{k}} - \mu)$$

Condensation energy

Going over continuum approximation,

$$\langle E \rangle_n - \langle E \rangle_s = \frac{1}{2} D(E_F) \Delta^2,$$

by letting $\mu = 0$ and assuming $\Delta \ll \hbar\omega_D$.

Recall that the condensation energy from thermodynamics:

$$\frac{1}{2} \mu_0 H_c^2$$

Therefore

$$H_c(0) = \sqrt{D(E_F)/\mu_0} \Delta(0) \quad (4)$$

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Canonical transformation

Canonical transformation

- Variational method (we have seen);
- Transform $\mathcal{H}$ into a **diagonalisable** form (we'll see):
 - Enable to obtain **excited states** & **finite temperature** properties.

A few steps ...

$$\mathcal{H} = \sum_{\mathbf{k}, \sigma} (\epsilon_{\mathbf{k}} - \mu) n_{\mathbf{k}\sigma} - |V| \sum_{\mathbf{k}\mathbf{k}'} c_{\mathbf{k}\uparrow}^{\dagger} c_{-\mathbf{k}\downarrow}^{\dagger} c_{-\mathbf{k}'\downarrow} c_{\mathbf{k}'\uparrow} \quad (5)$$

One may write

$$c_{-\mathbf{k}\downarrow} c_{\mathbf{k}\uparrow} = \langle c_{-\mathbf{k}\downarrow} c_{\mathbf{k}\uparrow} \rangle + \underbrace{(c_{-\mathbf{k}\downarrow} c_{\mathbf{k}\uparrow} - \langle c_{-\mathbf{k}\downarrow} c_{\mathbf{k}\uparrow} \rangle)}_{\text{small}} \quad (6)$$

Mean field Hamiltonian

One can approximate

$$c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger c_{-k'\downarrow} c_{k'\uparrow} \approx c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger \langle c_{-k'\downarrow} c_{k'\uparrow} \rangle + \langle c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger \rangle c_{-k'\downarrow} c_{k'\uparrow} \quad (7)$$

$$- \langle c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger \rangle \langle c_{-k'\downarrow} c_{k'\uparrow} \rangle \quad (8)$$

$$\mathcal{H}_{\text{MF}} = \sum_{k,\sigma} (\epsilon_k - \mu) c_{k\sigma}^\dagger c_{k\sigma} \quad (9)$$

$$- |V| \sum_{kk'} \left(c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger \langle c_{-k'\downarrow} c_{k'\uparrow} \rangle + \langle c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger \rangle c_{-k'\downarrow} c_{k'\uparrow} - \langle c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger \rangle \langle c_{-k'\downarrow} c_{k'\uparrow} \rangle \right) \quad (10)$$

Use

$$\Delta \equiv |V| \sum \langle c_{-k\downarrow} c_{k\uparrow} \rangle \quad (11)$$

to rewrite as

$$\mathcal{H}_{\text{MF}} = \sum_{k,\sigma} (\epsilon_k - \mu) c_{k\sigma}^\dagger c_{k\sigma} - \sum_k \left(\Delta c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger + \Delta^* c_{-k\downarrow} c_{k\uparrow} - \Delta \langle c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger \rangle \right). \quad (12)$$

Diagonalised

$$\mathcal{H}_{\text{MF}} = \sum_{\mathbf{k}, \sigma} (\epsilon_{\mathbf{k}} - \mu) c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}\sigma} - \sum_{\mathbf{k}} \left(\Delta c_{\mathbf{k}\uparrow}^\dagger c_{-\mathbf{k}\downarrow}^\dagger + \Delta^* c_{-\mathbf{k}\downarrow} c_{\mathbf{k}\uparrow} - \Delta \langle c_{\mathbf{k}\uparrow}^\dagger c_{-\mathbf{k}\downarrow}^\dagger \rangle \right).$$

Much alike defining the ladder operators for harmonic oscillator problem, we'd like to rewrite

$\mathcal{H} = \text{const.} + \text{counting number of whatever new quanta (particles)}$

Here's the magic (Bogoliubov-Valatin transformation)

$$\gamma_{\mathbf{k}\uparrow}^\dagger = u_{\mathbf{k}}^* c_{\mathbf{k}\uparrow}^\dagger - v_{\mathbf{k}}^* c_{-\mathbf{k}\downarrow} \quad (13)$$

$$\gamma_{\mathbf{k}\downarrow}^\dagger = u_{\mathbf{k}}^* c_{-\mathbf{k}}^\dagger + v_{\mathbf{k}}^* c_{\mathbf{k}\uparrow}, \quad (14)$$

Direct substitution ... lengthy, but

Coefficients

One can show

$$\mathcal{H}_{\text{MF}} = \sum_{\mathbf{k}} \left((\epsilon_{\mathbf{k}} - \mu) - E_{\mathbf{k}} + \Delta \langle c_{\mathbf{k}\uparrow}^\dagger c_{-\mathbf{k}\downarrow}^\dagger \rangle \right) + \sum_{\mathbf{k}} E_{\mathbf{k}} \left(\gamma_{\mathbf{k}\uparrow}^\dagger \gamma_{\mathbf{k}\uparrow} + \gamma_{\mathbf{k}\downarrow}^\dagger \gamma_{\mathbf{k}\downarrow} \right) \quad (15)$$

if

$$2(\epsilon_{\mathbf{k}} - \mu)u_{\mathbf{k}}v_{\mathbf{k}} + \Delta^* v_{\mathbf{k}}^2 - \Delta u_{\mathbf{k}}^2 = 0. \quad (16)$$

The latter condition, as we define

$$E_{\mathbf{k}} = \sqrt{(\epsilon_{\mathbf{k}} - \mu)^2 + |\Delta|^2}, \quad (17)$$

can be shown to be equivalent to

$$|v_{\mathbf{k}}|^2 = \frac{1}{2} \left(1 - \frac{(\epsilon_{\mathbf{k}} - \mu)}{E_{\mathbf{k}}} \right) \quad (18)$$

Energy

$$\mathcal{H}_{\text{MF}} = \underbrace{\sum_{\mathbf{k}} \left((\epsilon_{\mathbf{k}} - \mu) - E_{\mathbf{k}} + \Delta \langle c_{\mathbf{k}\uparrow}^\dagger c_{-\mathbf{k}\downarrow}^\dagger \rangle \right)}_{\text{GS energy}} + \underbrace{\sum_{\mathbf{k}} E_{\mathbf{k}} \left(\gamma_{\mathbf{k}\uparrow}^\dagger \gamma_{\mathbf{k}\uparrow} + \gamma_{\mathbf{k}\downarrow}^\dagger \gamma_{\mathbf{k}\downarrow} \right)}_{\text{collection of new particles; bogoliubons}} \quad (19)$$

- $E_{\mathbf{k}}$ the excitation energy
- Δ the energy gap

Finite temperature

Finite temperature gap

$$\Delta \equiv |V| \sum \langle c_{-k\downarrow} c_{k\uparrow} \rangle \quad (20)$$

can be rewritten as

$$\Delta = |V| \sum_k u_k^* v_k \left(1 - \gamma_{k\uparrow}^\dagger \gamma_{k\uparrow} - \gamma_{k\downarrow}^\dagger \gamma_{k\downarrow} \right) \quad (21)$$

$$= |V| \sum_k u_k v_k^* (1 - 2f(E_k)) \quad (22)$$

$$= |V| \sum_k \frac{\Delta}{2E_k} (1 - 2f(E_k)) \quad (23)$$

since γ s are fermion operators,

$$\langle \gamma_{k\uparrow}^\dagger \gamma_{k\uparrow} \rangle = f(E_k), \quad (24)$$

where $f(E) = 1/(e^{E/k_B T} + 1)$.

$\Delta(0)$ and T_c

The gap equation,

$$\frac{1}{V} = \frac{1}{2} \sum_k \frac{1}{E_k} \tanh \frac{E_k}{2k_B T}. \quad (25)$$

To get T_c where $\Delta = 0$, change the sum to an integral,

$$\frac{1}{D(E_F)|V|} = \int_0^{\hbar\omega_D/2k_B T_c} \frac{dx \tanh x}{x} \quad (26)$$

$$= \ln(1.13\hbar\omega_D/k_B T_c). \quad (27)$$

$$k_B T_c = 1.13\hbar\omega_D e^{-1/D(E_F)|V|} \quad (28)$$

Comparing with

$$\Delta(0) = 2\hbar\omega_D e^{-1/D(E_F)V} \quad (29)$$

gives

$$2\Delta(0) = 3.52k_B T_c. \quad (30)$$

Tunneling: normal to
superconductor

Tunneling

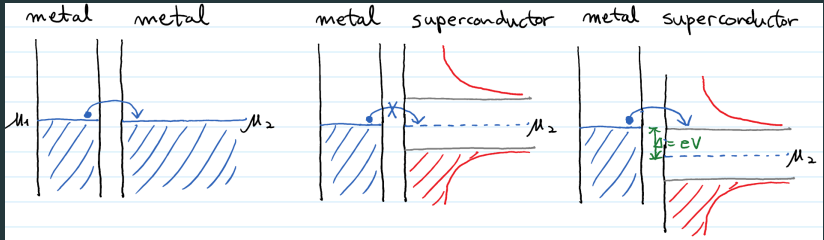


Figure 1: Energy versus density of states for three different contacts of metal-metal, metal-superconductor and superconductor at zero bias, and metal-superconductor at threshold bias.

Density of states

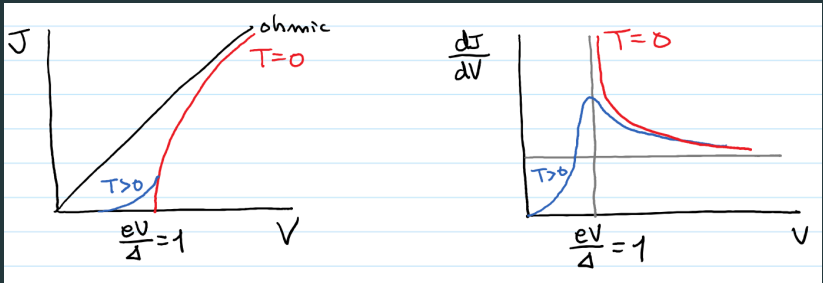


Figure 2: Current-voltage characteristic (left) and differential conductance versus voltage (right).

Tunneling: superconductor to
superconductor

Different gaps ($T = 0$)

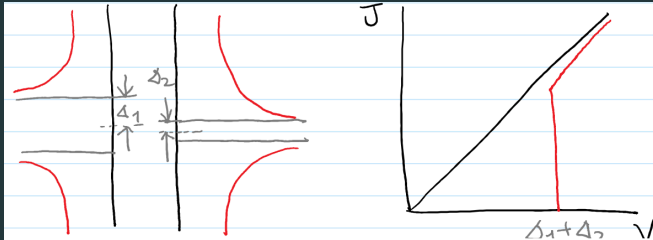


Figure 3: Density of states (left) and I-V characteristic of two different superconductors at zero temperature.

Different gaps ($T \neq 0$)

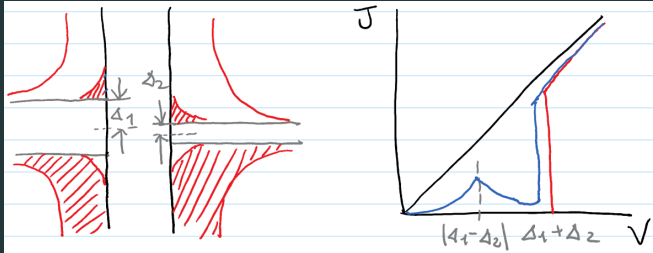


Figure 4: Density of states (left) and I-V characteristic of two different superconductors at finite temperature.